KERALA POLYTECHNIC · REVISION 2026

Practicum | Semester I | 2.5 Credits | 60 Contact Hours | 1:0:3:0 (L:T:P:R)

Managerial Economics

Course Code 1143 · Commercial Practice

PROGRAMME **Commercial Practice**	SEMESTER **I**
CREDITS **2.5**	CONTACT HOURS **60**
TEACHING SCHEME **1:0:3:0**	CIE : ESE **40 : 60**

📋 Official Source Declaration SYLLABUS INTEGRITY

This handbook is derived from the official Kerala Polytechnic Revision 2026 syllabus for **Course 1143 — Managerial Economics**.

✅ **Verified:** All modules, topics, subtopics, learning outcomes, teaching schemes, assessment patterns, and self-learning activities have been extracted directly from the official syllabus PDF.

⚠️ **Source inconsistency disclosed:** The PDF lists **Module IV** with 3 instructional hours (L) and 8 practical hours (P), totalling 11 hours, but the contact hours summary states 60 total hours across 4 modules. The per-module breakdown (3+10, 5+13, 4+14, 3+8 = 60) reconciles correctly. The handbook treats the module-level hour figures as authoritative.

Last verified: July 2026 · All elaborations are educational and not sourced from the syllabus unless explicitly attributed.

📊 Course Dashboard AT A GLANCE

📘 **Programme**
Commercial Practice

📚 **Course Type**
Practicum · 60 contact hours

🎯 **Credits**
2.5 (1L + 3P + 6 self-learning/week)

📝 **CIE**
40 marks (attendance + practicals + CAs)

📝 **ESE**
60 marks (2.5 hrs theory exam)

🧪 **Self-Learning**
48 hours total (6 hrs/week × 8 weeks)

Teaching and Assessment Scheme

Teaching/Week	Self-learning/Week	Total Hours/Sem	Credits	CIE	ESE	Total
10	6	60	2.5	40	60	100

L: Lecture (1 hr, 1 credit) · **T:** Tutorial (1 hr, 1 credit) · **P:** Practical (1 hr, 0.5 credit) · **R:** Remedial (1 hr, 0.5 credit)

🧭 How to Use This Handbook STUDY PATH

1

Understand

Read each module's core concepts, definitions, and principles.

2

Visualise

Study the diagrams, graphs, and tables to see economic relationships.

3

Apply

Work through the examples, calculators, and self-learning activities.

4

Revise

Use the formula bank, Q&A, model paper, and revision cards to prepare for exams.

🎯 Course Outcomes WHAT YOU WILL LEARN

By the end of this course, you will be able to:

CO1 · Apply

Apply the fundamental concepts of Managerial Economics in decision-making.

Bloom: Apply

CO2 · Demonstrate

Demonstrate the concepts of demand and supply, and compare the law of demand with the law of supply.

Bloom: Apply

CO3 · Illustrate

Illustrate production theory and economies of scale.

Bloom: Apply

CO4 · Classify

Classify various market forms and pricing methods.

Bloom: Apply

CO–PO Mapping

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	-	-	1	2	-	-	1	-	1
CO2	2	2	-	-	-	1	-	-	1	-	1
CO3	2	2	-	-	-	2	-	-	1	-	1
CO4	2	3	-	-	-	2	-	-	1	-	1

Legends: 1 — Low · 2 — Medium · 3 — High · - — No Correlation

🗺️ Curriculum Coverage Map MODULE ROADMAP

Module–Topic–Hours–CO Mapping

Module	Title	L	P	CO	Bloom
I	Introduction to Managerial Economics	3	10	CO1	Apply
II	Theory of Demand and Supply	5	13	CO2	Apply
III	Concepts of Production	4	14	CO3	Apply
IV	Forms of Market	3	8	CO4	Apply

Total: 15 lecture hours + 45 practical hours = 60 contact hours.

🔗 Module Connection Roadmap HOW IDEAS FLOW

📘
M1: Introduction
Basic concepts · Utility · Law of DMU
Foundation

📗
M2: Demand & Supply
Demand · Supply · Elasticity
Market forces

📙
M3: Production

📕
M4: Market Forms

Production function · Costs · Returns to scale

Firm behaviour

Perfect competition · Monopoly · Pricing

Market outcomes

MODULE I

Introduction to Managerial Economics

📚 3L + 10P = 13 hrs | CO1 · Apply | 📘 Core concepts

This module builds the foundation for understanding how economic principles apply to business decision-making. You will learn about human wants, utility, value, price, and the law of diminishing marginal utility.

▶ **Basic concepts of Economics and Managerial Economics**

What is Economics?

Economics is the social science that studies how individuals, businesses, governments, and societies make choices about how to allocate scarce resources to satisfy unlimited wants. It is broadly divided into two branches:

- **Microeconomics:** Studies the behaviour of individual consumers, firms, and markets. Focuses on price determination, demand, supply, and production decisions at the firm level.
- **Macroeconomics:** Studies the economy as a whole — national income, employment, inflation, economic growth, and fiscal/monetary policy.

What is Managerial Economics?

Managerial Economics is the application of economic theory and quantitative methods to business decision-making. It bridges the gap between abstract economic theory and practical business management.
It helps managers:

- Make optimal decisions about resource allocation
- Analyse demand and forecast sales
- Determine pricing strategies
- Evaluate production and cost structures
- Understand market competition and strategic behaviour

Definitions of Economics

Several schools of thought have defined economics differently:

- **Wealth Definition (Adam Smith):** Economics is the science of wealth — how nations produce and distribute wealth. (1776, *An Inquiry into the Nature and Causes of the Wealth of Nations*)
- **Welfare Definition (Alfred Marshall):** Economics is the study of mankind in the ordinary business of life — it examines how people earn and use income, and how this affects human welfare. (1890, *Principles of Economics*)
- **Scarcity Definition (Lionel Robbins):** Economics is the science that studies human behaviour as a relationship between ends and scarce means that have alternative uses. (1932, *An Essay on the Nature and Significance of Economic Science*)
- **Growth Definition (Paul Samuelson):** Economics is the study of how societies use scarce resources to produce valuable commodities and distribute them among different people, with a focus on economic growth over time.

> Managerial Economics = Economic Theory + Business Decision-Making

Malayalam support: [illegible] (Economics) [illegible].
[illegible].

💡 **Key takeaway:** Managerial economics is *not* just theory — it is a practical toolkit for making better business decisions using economic logic.

▶ **Human wants, Wealth and Welfare, Value and Price**

Human Wants

Human wants are the desires for goods and services that provide satisfaction. They are:

- **Unlimited:** As one want is satisfied, new wants emerge.
- **Recurring:** Many wants (like food, clothing) arise repeatedly.
- **Competitive:** Wants compete for limited resources — you must choose which to satisfy.
- **Complementary:** Some wants go together (e.g., car + fuel).
- **Varying in intensity:** Some wants are urgent (basic needs), others are less pressing.

Goods are the objects that satisfy human wants. They can be:

- **Free goods:** Available without cost (air, sunlight).
- **Economic goods:** Scarce and require effort to obtain (food, clothing, machinery).
- **Consumer goods:** Used directly by consumers (bread, mobile phones).
- **Producer goods:** Used to produce other goods (machinery, raw materials).

Wealth and Welfare

- **Wealth:** The stock of all goods, assets, and resources that have economic value. It includes physical assets (land, buildings, machinery) and financial assets (money, stocks, bonds).
- **Welfare:** The well-being or standard of living of individuals and society. It is broader than wealth — it includes health, education, happiness, and quality of life.

Classical economists focused on wealth creation; modern economists emphasise welfare and human development.

Value and Price

- **Value:** The worth of a good or service, often expressed in terms of its utility or exchange value. "*Value in use*" (utility) vs. "*value in exchange*" (market price).
- **Price:** The monetary amount at which a good or service is exchanged in the market. Price is the *expression of value in money terms*.

For example, water has high *value in use* (essential for life) but low price (cheap), while diamonds have low *value in use* but high price (expensive). This is the famous **"diamond-water paradox"** that economists explain through scarcity and marginal utility.

> **Malayalam support:** [illegible] [illegible] (wants) [illegible]. [illegible] (wealth) [illegible] [illegible] [illegible] [illegible]. [illegible] (price) [illegible] [illegible]

[illegible] [illegible], [illegible] (value) [illegible] [illegible] [illegible]
[illegible].

▸ **Concept of Utility and Law of Diminishing Marginal Utility**

Concept of Utility

Utility is the satisfaction, pleasure, or benefit that a consumer derives from consuming a good or service. It is the foundation of demand theory.

- **Total Utility (TU):** The total satisfaction obtained from consuming a certain quantity of a good.
- **Marginal Utility (MU):** The additional satisfaction gained from consuming one more unit of a good. $MU = \Delta TU / \Delta Q$.

Types of Utility:

- **Cardinal Utility:** Assumes utility can be measured in numerical units (utils).
- **Ordinal Utility:** Assumes utility can only be ranked or ordered (preference ordering), not measured absolutely.
- **Total Utility:** Overall satisfaction from all units consumed.
- **Marginal Utility:** Satisfaction from the last unit consumed.

Law of Diminishing Marginal Utility (LDMU)

The law states: "*As a consumer consumes more and more units of a good, the marginal utility derived from each additional unit eventually declines.*"
This law applies under the assumption that:

- The consumer's taste/preferences remain unchanged.
- Units of the good are homogeneous (identical).
- Consumption is continuous without significant breaks.
- There are no changes in the consumer's income or the prices of substitutes/complements.

Example

Diminishing Marginal Utility — Ice Cream

Units Consumed	Total Utility (TU)	Marginal Utility (MU)
1	30	30
2	50	20
3	62	12
4	68	6
5	70	2
6	70	0
7	68	-2

After the 6th unit, MU becomes zero, and after the 7th, it becomes negative (disutility).

Importance in Managerial Economics

- Explains the downward-sloping demand curve (consumers buy more only if price falls).
- Helps in pricing decisions — firms know that each additional unit sold yields lower marginal utility, so price must decrease.
- Used in consumer behaviour analysis and market segmentation.

📝 **WORKED EXAMPLE**

Given: A consumer's TU from consuming 1 to 5 chocolates is: 10, 18, 24, 28, 30.

Required: Calculate MU for each unit and identify the point where MU starts diminishing.

Calculation:

$MU_1 = 10 – 0 = 10$

$MU_2 = 18 – 10 = 8$

$MU_3 = 24 – 18 = 6$

$MU_4 = 28 – 24 = 4$

$MU_5 = 30 – 28 = 2$

Answer: MU declines from the very first additional unit (from 10 → 8 → 6 → 4 → 2), illustrating the law.

Malayalam support: [illegible], [illegible] (marginal utility) [illegible]. [illegible].

💡 **Business application:** This law explains why businesses offer discounts on bulk purchases — the marginal value to the consumer declines, so a lower price per unit is needed to sell more.

▸ **Standard of Living**

Concept of Standard of Living

Standard of living refers to the level of wealth, comfort, material goods, and necessities available to a person or community. It is a measure of the quality of life and economic well-being.
Indicators include:

- **Income and wealth:** Higher income enables access to better goods and services.
- **Access to education and healthcare:** Essential for long-term well-being.
- **Housing and infrastructure:** Quality of housing, transport, and utilities.
- **Leisure and recreation:** Time and resources for rest and enjoyment.
- **Environmental quality:** Clean air, water, and safe surroundings.

Standard of living is closely related to **productivity** — a nation's ability to produce goods and services efficiently determines its average standard of living.

> 💡 **In managerial economics:** Understanding standard of living helps managers assess market demand — as living standards rise, consumers demand higher-quality goods and services, influencing product strategies.

> **Malayalam support:** □□□□□ □□□□□□□□ (standard of living) □□□□□□ □□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□. □□□ □□□□□□□, □□□□□□□□□□□, □□□□□□□, □□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□.

> 📘 **Module I summary:** Managerial economics applies economic principles to business decisions. Human wants are unlimited, and utility (satisfaction) diminishes with each additional unit consumed — a key insight for pricing and demand analysis.

MODULE II

Theory of Demand and Supply

📚 5L + 13P = 18 hrs | CO2 · Apply | 📗 Market forces

This module covers the foundational concepts of demand and supply — the two forces that determine market prices and quantities. You will learn about the laws of demand and supply, elasticity, and their practical applications.

▸ **Demand — Meaning, Types, and Factors Affecting Demand**

Meaning of Demand

Demand is the quantity of a good or service that consumers are *willing and able* to purchase at various prices during a given period of time. Demand is always expressed with reference to:

- **Price:** The amount the consumer must pay.
- **Time:** The period over which demand is measured.
- **Quantity:** The amount demanded at a given price.

> Demand = Desire + Ability to Pay + Willingness to Buy

Types of Demand

- **Individual demand:** Demand by a single consumer.
- **Market demand:** Total demand by all consumers in the market (sum of individual demands).
- **Direct demand:** Demand for final consumer goods (e.g., food, clothing).
- **Derived demand:** Demand for goods that are used in the production of other goods (e.g., demand for steel is derived from demand for cars).
- **Joint demand:** Demand for goods that are used together (e.g., car and fuel, printer and ink).
- **Composite demand:** Demand for a good that has multiple uses (e.g., electricity for lighting, heating, and power).

Factors Affecting Demand

- **Price of the good (P):** Inverse relationship — higher price, lower demand (ceteris paribus).
- **Income of consumers (Y):** Normal goods: demand ↑ as income ↑. Inferior goods: demand ↓ as income ↑.
- **Prices of related goods:**
  - **Substitutes:** If price of substitute rises, demand for the good rises (e.g., tea and coffee).
  - **Complements:** If price of complement rises, demand for the good falls (e.g., car and fuel).
- **Tastes and preferences:** Changes in consumer tastes shift demand.
- **Expectations:** Future price/income expectations affect current demand.
- **Population:** More consumers → higher market demand.
- **Advertising and branding:** Effective promotion can increase demand.

> **Malayalam support:** □□□□□□□□ (demand) □□□□□□□ □□□□ □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□. □□□□, □□□□□□□□, □□□□□□□□□□□□□ □□□□□□□□□□□□□□□ □□□□,

▸ Law of Demand and Exceptions

Law of Demand

The **law of demand** states: "*Other things being equal (ceteris paribus), the quantity demanded of a good falls when its price rises, and rises when its price falls.*"

- Demand curve slopes **downward** from left to right.
- It reflects the **inverse relationship** between price and quantity demanded.

Reasons for the Downward-Sloping Demand Curve

- **Law of Diminishing Marginal Utility:** As consumption increases, MU declines, so consumers are willing to pay less for additional units.
- **Income effect:** When price falls, the consumer's real income rises, enabling them to buy more.
- **Substitution effect:** When price falls, the good becomes relatively cheaper than substitutes, so consumers switch to it.
- **New consumers:** Lower prices attract new buyers who were previously unable or unwilling to purchase.

Exceptions to the Law of Demand

- **Giffen goods:** Inferior goods for which demand increases as price rises (e.g., staple foods in poor economies). Named after Sir Robert Giffen.
- **Veblen goods (status goods):** Goods where higher price signals higher status, increasing demand (e.g., luxury watches, designer handbags).
- **Speculative goods:** Goods that people buy expecting future price rises (e.g., stocks, real estate).
- **Essential goods:** Goods that people must buy regardless of price (e.g., life-saving medicines).

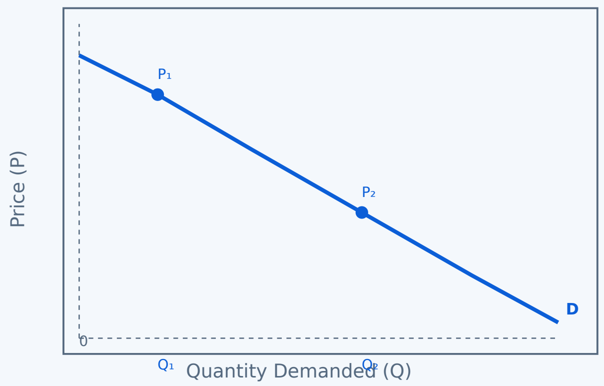

Fig. 1: The downward-sloping demand curve shows that as price falls from P_1 to P_2, quantity demanded increases from Q_1 to Q_2.

Malayalam support: [illegible] — [illegible], [illegible]. [illegible] Giffen [illegible], Veblen [illegible].

▸ **Elasticity of Demand**

Concept of Elasticity of Demand

Elasticity of demand measures the responsiveness of quantity demanded to a change in one of its determinants (price, income, or price of related goods).

Types of Elasticity of Demand

- **Price Elasticity of Demand (Ed):** Measures responsiveness of Qd to a change in price.

  > Ed = (% Change in Quantity Demanded) / (% Change in Price)

- **Income Elasticity of Demand (Ey):** Measures responsiveness of Qd to a change in consumer income.

  > Ey = (% Change in Qd) / (% Change in Income)

- **Cross Elasticity of Demand (Exy):** Measures responsiveness of Qd of good X to a change in the price of good Y.

  > Exy = (% Change in Qd of X) / (% Change in Price of Y)

Types of Price Elasticity

- **Perfectly elastic (Ed = ∞):** Any price change → quantity demanded drops to zero. (Horizontal demand curve.)
- **Perfectly inelastic (Ed = 0):** Quantity demanded does not change with price. (Vertical demand curve; e.g., life-saving drugs.)
- **Relatively elastic (Ed > 1):** % change in Qd > % change in price. (Luxury goods.)
- **Relatively inelastic (Ed < 1):** % change in Qd < % change in price. (Necessities.)
- **Unit elastic (Ed = 1):** % change in Qd = % change in price.

Factors Affecting Elasticity of Demand

- **Nature of the good:** Necessities are inelastic; luxuries are elastic.
- **Availability of substitutes:** More substitutes → more elastic.
- **Proportion of income spent:** Higher proportion → more elastic.
- **Time period:** Demand is more elastic in the long run.
- **Addiction/habit:** Addictive goods are inelastic.

🧮 Price Elasticity of Demand Calculator

Initial Price (P_1) New Price (P_2) Initial Quantity (Q_1)

100	80	50

New Quantity (Q_2)

70

Enter values and click Calculate.

📝 **WORKED EXAMPLE**

Given: Price of a product rises from ₹50 to ₹60. Quantity demanded falls from 200 to 160 units.

Required: Calculate the price elasticity of demand and interpret.

Calculation:

% ΔP = (60 – 50)/50 × 100 = 20%

% ΔQ = (160 – 200)/200 × 100 = –20%

Ed = –20% / 20% = –1 (absolute value = 1)

Answer: Ed = 1 (unit elastic) — the percentage change in quantity demanded equals the percentage change in price.

Malayalam support: □□□□□□□□□□□ □□□□□□□□□ (elasticity) □□□□□□ □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□ □□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□ □□□□□ □□□□□□□□□□. □□□ □□□□□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□.

💡 **Business application:** Knowing price elasticity helps firms set optimal prices. If demand is elastic, lowering price increases total revenue. If inelastic, raising price increases total revenue.

▸ **Law of Supply and Elasticity of Supply**

Law of Supply

The **law of supply** states: "*Other things being equal, the quantity supplied of a good rises when its price rises, and falls when its price falls.*"

- Supply curve slopes **upward** from left to right.
- It reflects the **direct relationship** between price and quantity supplied.

Exceptions to the Law of Supply

- **Agricultural products:** In the short run, supply is fixed (e.g., a harvest is already grown).
- **Perishable goods:** Sellers may supply more even at falling prices to avoid wastage.
- **Labour supply:** Higher wages may eventually lead to reduced work hours (backward-bending supply curve).
- **Future expectations:** If sellers expect prices to rise further, they may withhold supply now.

Elasticity of Supply

Elasticity of supply (Es) measures the responsiveness of quantity supplied to a change in price.

> Es = (% Change in Quantity Supplied) / (% Change in Price)

Types:

- **Perfectly elastic (Es = ∞):** Any price change → infinite change in supply.
- **Perfectly inelastic (Es = 0):** Supply is fixed regardless of price (e.g., land).
- **Relatively elastic (Es > 1):** Supply responds more than proportionally.
- **Relatively inelastic (Es < 1):** Supply responds less than proportionally.
- **Unit elastic (Es = 1):** Supply responds proportionally.

Factors Affecting Elasticity of Supply

- **Time period:** Supply is more elastic in the long run (firms can adjust capacity).
- **Availability of inputs:** If inputs are readily available, supply is more elastic.
- **Production capacity:** Firms with idle capacity have more elastic supply.
- **Storage costs:** High storage costs make supply more elastic in the short run.

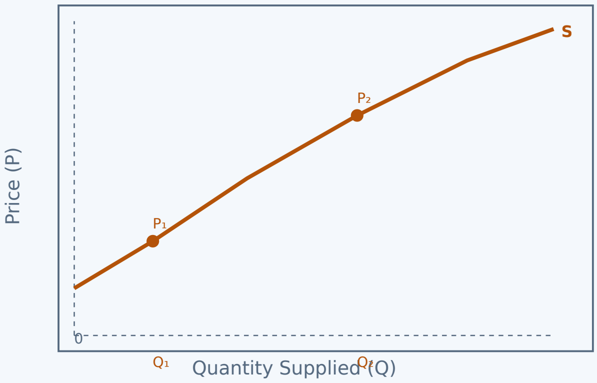

Fig. 2: The upward-sloping supply curve shows that as price rises from P_1 to P_2, quantity supplied increases from Q_1 to Q_2.

📝 **WORKED EXAMPLE**

Given: Price of a product rises from ₹80 to ₹100, and quantity supplied rises from 400 to 500 units.

Required: Calculate the elasticity of supply and interpret.

Calculation:

$\% \Delta P = (100 - 80)/80 \times 100 = 25\%$

$\% \Delta Q = (500 - 400)/400 \times 100 = 25\%$

$Es = 25\% / 25\% = 1$

Answer: Es = 1 (unit elastic) — supply changes proportionally with price.

Malayalam support: □□□□□ □□□□□ (law of supply) □□□□□□□□ — □□□ □□□□□□□□□ □□□□□□ □□□□□, □□□ □□□□□□□□□□ □□□□□□ □□□□□□. □□□□□ □□□□□□□□□ (elasticity of supply) □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□.

📗 **Module II summary:** Demand and supply are the twin forces that drive market prices. The law of demand (inverse relationship) and law of supply (direct relationship) together determine equilibrium price. Elasticity measures responsiveness — crucial for pricing and revenue decisions.

MODULE III

Concepts of Production

📚 4L + 14P = 18 hrs · CO3 · Apply · 📙 Firm behaviour

This module explores how goods and services are produced. You will learn about production functions, the factors of production, laws of returns, cost structures, and economies of scale.

▶ Production, Production Function, and Law of Variable Proportion

Meaning of Production

Production is the process of transforming inputs (factors of production) into outputs (goods and services) that satisfy human wants. It is the creation of utility.

- **Form utility:** Changing raw materials into finished goods.
- **Place utility:** Moving goods to where they are needed.
- **Time utility:** Making goods available when they are needed.

Production Function

The **production function** shows the maximum output that can be produced from a given set of inputs, using the best available technology.

> Q = f (L, K, Land, Entrepreneurship)

Where:

- **Q** = Quantity of output
- **L** = Labour
- **K** = Capital
- **Land** = Natural resources
- **Entrepreneurship** = Organisation and risk-taking

Law of Variable Proportion

The **Law of Variable Proportion** (also called the Law of Diminishing Returns) states that as more and more units of a variable input (e.g., labour) are added to fixed inputs (e.g., land, capital), the marginal product of the variable input eventually declines. Three stages:

- **Stage I (Increasing returns):** MP rises as fixed inputs are underutilised. Total product increases at an increasing rate.
- **Stage II (Diminishing returns):** MP falls but remains positive. Total product increases at a decreasing rate. This is the *rational zone* for production.
- **Stage III (Negative returns):** MP becomes negative. Total product starts to fall. This stage is *irrational*.

Law of Variable Proportion — Example

Units of Labour (L)	Total Product (TP)	Marginal Product (MP)	Stage
1	10	10	I (Increasing)
2	25	15	
3	45	20	
4	60	15	II (Diminishing)

Units of Labour (L)	Total Product (TP)	Marginal Product (MP)	Stage
5	70	10	
6	75	5	
7	75	0	III (Negative)
8	70	−5	

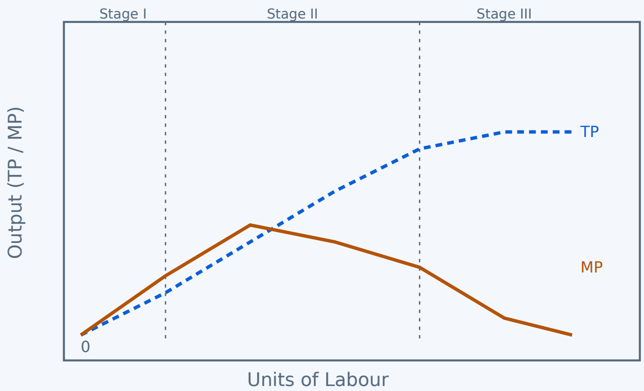

Fig. 3: The Law of Variable Proportion — Stage I (increasing returns), Stage II (diminishing returns), Stage III (negative returns).

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▸ Law of Returns to Scale and Factors of Production

Law of Returns to Scale

The **Law of Returns to Scale** examines the effect on output when *all* inputs are increased proportionally in the long run (when all factors are variable).

- **Increasing returns to scale:** Output increases by a greater proportion than the increase in inputs. (e.g., inputs × 2 → output × 2.5). Occurs due to specialisation and economies of scale.
- **Constant returns to scale:** Output increases in the same proportion as inputs. (e.g., inputs × 2 → output × 2).
- **Decreasing returns to scale:** Output increases by a smaller proportion than the increase in inputs. (e.g., inputs × 2 → output × 1.5). Occurs due to managerial diseconomies.

> $Q = f(\lambda L, \lambda K, \lambda Land) \rightarrow$ If $\lambda Q = \lambda \times Q$: Constant; if $\lambda Q > \lambda Q$: Increasing; if $\lambda Q < \lambda Q$: Decreasing.

Difference: Law of Variable Proportion vs. Law of Returns to Scale

Aspect	Law of Variable Proportion	Law of Returns to Scale
Time frame	Short run	Long run
Inputs	One variable, others fixed	All inputs variable
Focus	Marginal product of one factor	Overall scale of production
Stages	Increasing, Diminishing, Negative	Increasing, Constant, Decreasing

Factors of Production

- **Land:** Natural resources used in production (land itself, minerals, water, forests). Its supply is fixed.
- **Labour:** Human effort — physical and mental — used in production. Labour is the most active factor.
- **Capital:** Man-made goods used in production (machinery, tools, buildings, raw materials). Capital is produced by labour and land.
- **Entrepreneurship:** The organising and risk-taking function. The entrepreneur combines the other factors, takes decisions, and bears uncertainty.

> **Malayalam support:** [illegible]. [illegible] (factors of production) — [illegible], [illegible], [illegible], [illegible] — [illegible].

▸ **Costs of Production — Types and Curves**

Types of Costs

- **Fixed Costs (FC):** Costs that do not change with the level of output (e.g., rent, salaries, insurance).
- **Variable Costs (VC):** Costs that change with the level of output (e.g., raw materials, wages of temporary workers, electricity for production).
- **Total Cost (TC):** Sum of fixed and variable costs. TC = FC + VC.
- **Average Fixed Cost (AFC):** FC divided by output. AFC = FC / Q. AFC declines as output increases.
- **Average Variable Cost (AVC):** VC divided by output. AVC = VC / Q. Typically U-shaped.
- **Average Total Cost (ATC):** TC divided by output. ATC = TC / Q = AFC + AVC. U-shaped.
- **Marginal Cost (MC):** The additional cost of producing one more unit. MC = ΔTC / ΔQ. U-shaped.

Cost Curves

- **Fixed Cost Curve:** Horizontal line at the level of fixed cost.
- **Variable Cost Curve:** Upward-sloping from the origin, initially at a decreasing rate, then increasing.
- **Total Cost Curve:** Parallel to the VC curve, starting from the FC level.
- **AFC Curve:** Downward-sloping, approaching zero as output increases (rectangular hyperbola).
- **AVC Curve:** U-shaped — falls initially, then rises after the minimum point.
- **ATC Curve:** U-shaped — lies above AVC, with the gap between them narrowing as output rises (since AFC declines).
- **MC Curve:** U-shaped — intersects AVC and ATC at their minimum points.

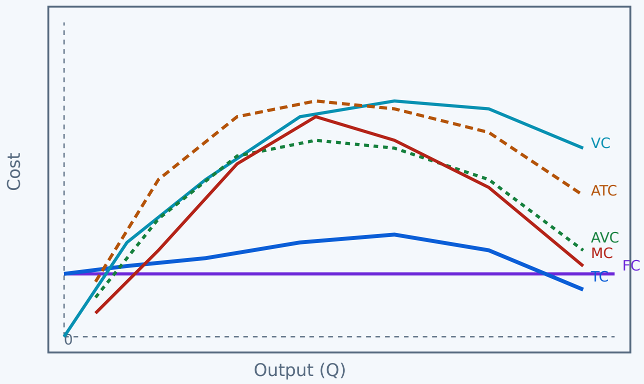

Fig. 4: Cost curves — FC (horizontal), VC (upward), TC (parallel to VC), ATC (U-shaped), AVC (U-shaped), MC (U-shaped, intersecting ATC and AVC at their minima).

💡 **Key insight:** The MC curve intersects the AVC and ATC curves at their lowest points. This is the "efficient scale" — the output level that minimises average cost.

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▸ **Scales of Production, Economies and Diseconomies of Scale**

Scales of Production

- **Small-scale production:** Low output, limited specialisation, higher average costs.
- **Medium-scale production:** Moderate output, some specialisation, lower average costs.
- **Large-scale production:** High output, extensive specialisation, lowest average costs (up to a point).

Economies of Scale

Economies of scale are the cost advantages that firms obtain as they expand production. Average costs fall as output increases.
Internal economies (within the firm):

- **Technical economies:** Better machinery, division of labour, research and development.
- **Managerial economies:** Specialised management, better organisation.
- **Financial economies:** Easier access to credit, lower interest rates.
- **Marketing economies:** Bulk purchasing and selling, advertising efficiencies.
- **Risk-bearing economies:** Diversification across products/markets.

External economies (outside the firm):

- Improved infrastructure, skilled labour pool, availability of suppliers, industry clusters.

Diseconomies of Scale

Diseconomies of scale occur when a firm becomes too large and average costs begin to rise.

- **Managerial diseconomies:** Communication breakdowns, bureaucracy, coordination problems.
- **Labour diseconomies:** Worker alienation, decreased productivity, industrial relations problems.
- **Financial diseconomies:** Higher interest rates for very large borrowings.
- **Technical diseconomies:** Inefficiencies due to excessively large plant size.

> 💡 **Business application:** Firms aim to operate at the scale where average costs are minimised — the "minimum efficient scale" (MES). Beyond that, diseconomies set in.

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📙 **Module III summary:** Production transforms inputs into outputs. The law of variable proportion (short run) and returns to scale (long run) explain how output responds to input changes. Cost curves help firms identify efficient production levels. Economies of scale drive cost reduction until diseconomies emerge.

MODULE IV Forms of Market

📚 3L + 8P = 11 hrs | CO4 · Classify | 📕 Market outcomes

This module examines the different market structures in which firms operate — from perfect competition to monopoly — and the pricing methods firms use in each.

▶ **Revenue Concept and Various Forms of Market**

Revenue Concepts

- **Total Revenue (TR):** The total income from selling a given quantity. TR = Price × Quantity (TR = P × Q).
- **Average Revenue (AR):** Revenue per unit sold. AR = TR / Q = Price (under normal conditions).
- **Marginal Revenue (MR):** The additional revenue from selling one more unit. MR = ΔTR / ΔQ.

$$AR = P \cdot MR = \Delta TR / \Delta Q$$

Forms of Market

Market structure is classified based on:

- **Number of firms** in the market
- **Nature of the product** (homogeneous or differentiated)
- **Barriers to entry** (how easy it is for new firms to enter)
- **Information availability** (perfect or imperfect)
- **Control over price** (price-taker or price-maker)

1. Perfect Competition

- **Many firms** — each is a price-taker.
- **Homogeneous product** — identical goods.
- **Free entry and exit** — no barriers.
- **Perfect information** — buyers and sellers know everything.
- **No control over price** — firms accept the market price.
- **Examples:** Agricultural markets (wheat, rice), stock markets (to some extent).

2. Monopoly

- **Single firm** — the only seller in the market.
- **No close substitutes** — unique product.
- **High barriers to entry** — legal, technical, or natural.
- **Price-maker** — the firm controls the price.
- **Types of monopoly:**
  - **Natural monopoly:** One firm can supply the entire market at lower cost than multiple firms (e.g., utilities).
  - **Legal monopoly:** Protected by patents, copyrights, or government franchise.
  - **Technological monopoly:** Based on proprietary technology.
  - **Geographic monopoly:** Only one seller in a region.
- **Examples:** Indian Railways (government monopoly), local electricity distribution.

3. Monopolistic Competition

- **Many firms** — each with a small market share.
- **Differentiated products** — goods are close substitutes but not identical.
- **Low barriers to entry** — relatively easy to enter.
- **Some control over price** — due to product differentiation.
- **Examples:** Restaurants, clothing brands, cosmetics, retail shops.

4. Oligopoly

- **Few large firms** — each has significant market power.
- **Interdependent decisions** — firms consider rivals' reactions.
- **High barriers to entry** — due to economies of scale, brand loyalty, etc.
- **May be homogeneous or differentiated** products.
- **Price rigidity** — firms are often reluctant to change prices (kinked demand curve).
- **Examples:** Automobile industry, mobile network operators, soft drinks (Coca-Cola vs. Pepsi).

5. Duopoly

- **Two firms** dominate the market.
- This is a special case of oligopoly.
- Firms may compete or collude.
- **Examples:** Airbus and Boeing in large aircraft manufacturing; in some local markets, two major retailers.

Comparison of Market Forms

Feature	Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly	Duopoly
Number of firms	Very many	One	Many	Few	Two
Product type	Homogeneous	Unique (no substitutes)	Differentiated	Homogeneous or differentiated	Homogeneous or differentiated
Barriers to entry	None	Very high	Low	High	High
Price control	Price-taker	Price-maker	Some	Significant	Significant
Interdependence	None	None	Some	High	High
Examples	Wheat, rice	Railways, utilities	Restaurants, clothing	Cars, telecom	Airbus & Boeing

Malayalam support: □□□□□□□□□□ □□□□□□□□ (market forms) — □□□□□□□ □□□□□□□ (perfect competition), □□□□□□□ (monopoly), □□□□□□□□□□ □□□□□□□ (monopolistic competition), □□□□□□□□□□ (oligopoly), □□□□□□□□□□□ (duopoly). □□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□□□.

▸ Various Methods of Pricing

Pricing Methods

- **Cost Plus Pricing (Mark-up Pricing):** Price is set by adding a standard mark-up to the cost of production. Price = Cost + (Mark-up % × Cost). Commonly used in retail, construction, and government contracts.
- **Target Pricing:** Price is set to achieve a specific target return on investment (ROI) or target profit. The firm determines the price that will generate the desired profit level at expected sales volume.
- **Marginal Cost Pricing:** Price is set equal to marginal cost. This is often used in competitive markets or to penetrate a new market. In the short run, a firm may price at MC to cover variable costs and contribute to fixed costs.
- **Going Rate Pricing (Competitive Pricing):** Price is set based on competitors' prices. The firm follows the industry average. Common in oligopolistic markets where price leadership exists.
- **Customary Pricing:** Price is set based on tradition or long-standing practice. Examples include a newspaper at ₹5, a cup of tea at ₹10. Prices are changed only when costs change significantly.
- **Differential Pricing (Price Discrimination):** Charging different prices to different customers for the same product, based on willingness to pay, time of purchase, quantity, or customer segment. Examples: student discounts, peak/off-peak pricing, airline tickets.

Summary of Pricing Methods

Method	Basis	Example
Cost Plus	Cost + Mark-up	Retail stores, government contracts
Target Pricing	Target ROI / profit	Manufacturers, project-based firms
Marginal Cost	Marginal cost of production	Short-run pricing, market entry
Going Rate	Competitors' prices	Oligopolistic markets, petrol prices
Customary	Tradition / long practice	Newspaper, tea, bus fare
Differential	Customer segment / time / quantity	Airline tickets, student discounts

💡 **Choosing a pricing method:** The choice depends on market structure, cost structure, competitive environment, and business objectives. In perfect competition, firms are price-takers; in monopoly, they are price-makers; in between, they have varying degrees of pricing freedom.

Malayalam support: □□□□□□□□□□ □□□□□□□ (pricing methods) — □□□□□ □□□□□□□□□□□□□□□□□□□□□ (cost plus), □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□ (target pricing), □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□ (going rate), □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□ (customary), □□□□□□□□□□ □□□ (differential pricing).

📕 **Module IV summary:** Market structures range from perfect competition (many firms, price-takers) to monopoly (one firm, price-maker). In between are monopolistic competition, oligopoly, and duopoly. Pricing methods vary — from cost-plus to differential pricing — depending on the market structure and business strategy.

📐 Formula Bank

QUICK REFERENCE

Total Utility

$$TU = \Sigma\ MU$$

Sum of marginal utilities from all units.

Marginal Utility

$$MU = \Delta TU / \Delta Q$$

Additional utility from one more unit.

Price Elasticity of Demand

$$Ed = (\%\Delta Qd) / (\%\Delta P)$$

Responsiveness of quantity demanded to price.

Income Elasticity of Demand

$$Ey = (\%\Delta Qd) / (\%\Delta Y)$$

Responsiveness of demand to income changes.

Cross Elasticity of Demand

$$Exy = (\%\Delta Qx) / (\%\Delta Py)$$

Responsiveness of demand for X to price of Y.

Elasticity of Supply

$$Es = (\%\Delta Qs) / (\%\Delta P)$$

Responsiveness of quantity supplied to price.

Production Function

$$Q = f(L, K, Land, E)$$

Output as a function of labour, capital, land, entrepreneurship.

Total Cost

$$TC = FC + VC$$

Fixed cost + variable cost.

Average Cost

$$ATC = TC / Q = AFC + AVC$$

Total cost per unit of output.

Marginal Cost

$$MC = \Delta TC / \Delta Q$$

Additional cost of one more unit.

Total Revenue

$$TR = P \times Q$$

Total income from sales.

Average Revenue

$$AR = TR / Q = P$$

Revenue per unit (price in perfect competition).

Marginal Revenue

$$MR = \Delta TR / \Delta Q$$

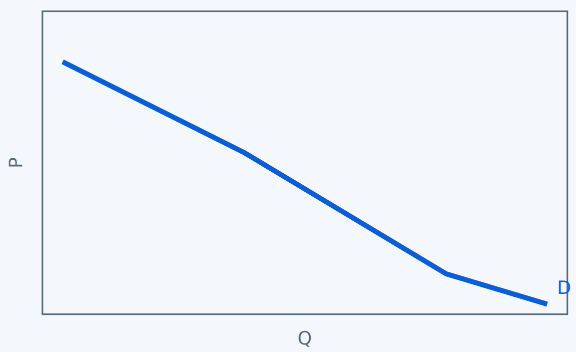
Additional revenue from one more unit.

Cost Plus Pricing

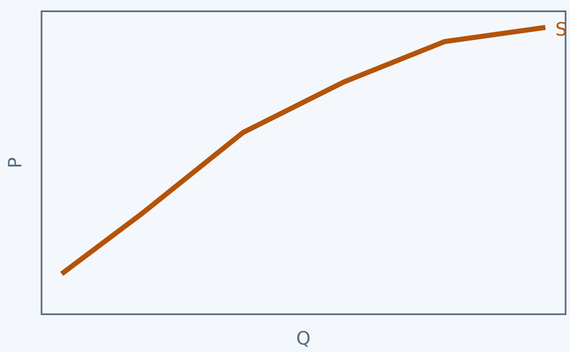
> Price = Cost + (Mark-up % × Cost)

Add a standard mark-up to cost.

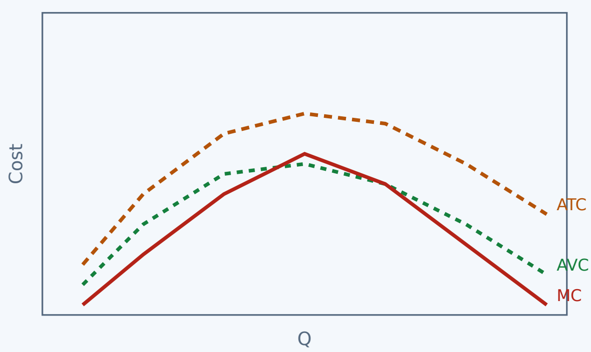
🖼️ Diagram Library for Rapid Revision VISUAL SUMMARY



Demand curve (downward sloping)



Supply curve (upward sloping)



Cost curves (ATC, AVC, MC)

🔗 See detailed diagrams in Module II, Module III.

🧪 Self-Learning Activities OFFICIAL TASKS

📹 **Module I — Week 1**
Watch an educational video on basic economic concepts and write a summary.
3 hrs

📝 **Module I — Week 2**
Identify an example from daily life that demonstrates the Law of Diminishing Marginal Utility and prepare a short report.

3 hrs

📋 **Module I — Week 3**
Prepare a list of personal wants and arrange them in order of priority.
3 hrs

📝 **Module I — Week 4**
Attempt unit-end review questions independently. Write 10 MCQ.
3 hrs

🏪 **Module II — Week 5**
Visit a local market, observe prices of different products, and prepare a table showing the price list of at least 10 items.
3 hrs

📝 **Module II — Week 6**
Write 10 examples of different types of demand in real life.
3 hrs

📦 **Module II — Week 7**
Observe real-life examples of the law of supply.
3 hrs

📊 **Module II — Week 8**
Compare the supply of two products with different elasticities of supply (e.g., vegetables vs. mobile phones).
3 hrs

🏭 **Module III — Week 9**
Choose a daily-life product, list the inputs used (land, labour, capital, entrepreneur).
3 hrs

📉 **Module III — Week 10**
Take a small business, classify expenses into Fixed Costs and Variable Costs.
3 hrs

📐 **Module III — Week 11**
Using imaginary data, draw Fixed Cost, Variable Cost, and Total Cost curves.
3 hrs

📝 **Module III — Week 12**
Attempt unit-end review questions independently. Write 10 MCQ.

3 hrs

🏢 Module IV — Week 13
Identify 5 products/services in the country where there is only one major supplier.

3 hrs

📊 Module IV — Week 14
Make a simple diagram to remember different market forms easily.

3 hrs

📝 Module IV — Week 15
Revise the module and prepare 10 questions.

3 hrs

Total self-learning: 15 weeks × 3 hrs = 45 hrs (as per syllabus).

📝 Assessment Methodology
CIE & ESE

Assessment Scheme — Practicum

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	ESE	Total
Duration	—	3 hrs	3 hrs	2 hrs	2 hrs	2.5 hrs	2.5 hrs	—
Exam marks	—	15	50	40	50	50	60	—
Converted to	5	10	10	10	10	10	—	—
Marks	5	10	10	10	10	10	60	100
Tentative schedule	End of sem	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	—

Practical (CIE) Evaluation Criteria

Scheme of Evaluation — Practical

Criteria	Description	Marks
Preparation & Participation	Attendance, punctuality, pre-class preparation, active participation	10
Practical Work / Activity Execution	Ability to perform assigned tasks such as drawing graphs, solving numerical problems, completing worksheets	10
Data Handling & Presentation	Neatness, accuracy in graphs, correct labelling, proper tabulation	5
Analysis & Interpretation	Ability to interpret graphs, explain economic relationships, apply laws, draw conclusions	10
Viva Voce / Conceptual Understanding	Understanding of concepts, ability to explain laws, relate theory with practical examples	10

Criteria	Description	Marks
Discipline & Teamwork	Classroom discipline, cooperation in group activities, attitude towards learning	5

Series Examination Pattern (Theory)

Series Test Pattern (2 hours)

Part	Type	Questions	Marks
Part A	Short answer (1 mark each)	4 × 2 modules = 8	8
Part B	Short essay (3 marks each)	2 × 2 modules = 4	12
Part C	Long answer with choice (7 marks each)	1 × 2 modules = 2	14
Total			**34**

End Semester Examination (ESE) Pattern

ESE Question Paper (2.5 hours, 60 marks)

Part	Type	Per module	Total Questions	Marks
Part A	1 mark each	2 questions	8	8
Part B	3 marks each (2 out of 3 per module)	2 questions	8	24
Part C	7 marks each (1 set with choice per module)	1 question	4	28
Total				**60**

❓ Module-wise Questions with Answers

EXAM PREPARATION

Module I — Introduction to Managerial Economics

Q: Define Managerial Economics. How does it help business decision-making?

A: Managerial Economics is the application of economic theory and quantitative methods to business decision-making. It helps managers by providing tools for demand analysis, cost analysis, pricing decisions, resource allocation, and strategic planning. It bridges the gap between abstract economic theory and practical business problems.

Q: State the Law of Diminishing Marginal Utility with an example.

A: The Law of Diminishing Marginal Utility states that as a consumer consumes more units of a good, the marginal utility from each additional unit eventually declines.

Example: Eating ice cream — the first scoop gives high satisfaction, the second less, and the fifth may give no additional satisfaction or even disutility.

Q: What are the different definitions of Economics?

A: Four major definitions: (1) Wealth definition (Adam Smith) — economics is the science of wealth. (2) Welfare definition (Alfred Marshall) — economics studies human welfare. (3) Scarcity definition (Lionel Robbins) — economics studies the allocation of scarce means to unlimited ends. (4) Growth definition (Paul Samuelson) — economics studies how societies use scarce resources to produce and distribute goods for growth and welfare.

Module II — Theory of Demand and Supply

Q: Explain the Law of Demand. What are its exceptions?

A: The Law of Demand states that, ceteris paribus, quantity demanded falls when price rises and rises when price falls. Exceptions include Giffen goods (demand increases with price), Veblen goods (status goods), speculative goods (expected future price rises), and essential goods (life-saving medicines).

Q: What is price elasticity of demand? How is it calculated?

A: Price elasticity of demand measures the responsiveness of quantity demanded to a change in price. It is calculated as: Ed = (% change in quantity demanded) / (% change in price). If $|Ed| > 1$, demand is elastic; if $|Ed| < 1$, inelastic; if $|Ed| = 1$, unit elastic.

Q: Distinguish between the Law of Demand and the Law of Supply.

A: The Law of Demand shows an inverse relationship between price and quantity demanded (downward-sloping curve). The Law of Supply shows a direct relationship between price and quantity supplied (upward-sloping curve). Demand is driven by consumers' willingness to pay; supply is driven by producers' willingness to sell.

Module III — Concepts of Production

Q: What is the Law of Variable Proportion? Explain its three stages.

A: The Law of Variable Proportion states that as more units of a variable input are added to fixed inputs, the marginal product eventually declines. Stages: I — Increasing returns (MP rises), II — Diminishing returns (MP falls but positive), III — Negative returns (MP negative). Stage II is the rational production zone.

Q: Differentiate between the Law of Variable Proportion and the Law of Returns to Scale.

A: The Law of Variable Proportion applies in the short run with one variable input and fixed others. The Law of Returns to Scale applies in the long run when all inputs are variable. The former focuses on marginal product of one factor; the latter focuses on the overall scale of production.

Q: What are economies of scale? Give examples of internal and external economies.

A: Economies of scale are cost advantages from larger production, leading to lower average costs. Internal economies: technical (better machinery), managerial (specialisation), financial (lower interest), marketing (bulk buying), risk-bearing (diversification). External economies: better infrastructure, skilled labour pool, industry clusters.

Module IV — Forms of Market

Q: Describe the features of perfect competition.

A: Perfect competition has: (1) many firms, each a price-taker; (2) homogeneous products; (3) free entry and exit; (4) perfect information; (5) no control over price. Examples: agricultural markets like wheat and rice.

Q: What are the different types of monopoly?

A: Types of monopoly: (1) Natural monopoly — one firm can supply the entire market at lower cost (utilities). (2) Legal monopoly — protected by patents, copyrights, or government franchise. (3) Technological monopoly — based on proprietary technology. (4) Geographic monopoly — only one seller in a region.

Q: Explain Cost Plus Pricing and Differential Pricing with examples.

A: Cost Plus Pricing: Setting price by adding a standard mark-up to cost (e.g., retail stores adding a 30% mark-up). Differential Pricing: Charging different prices to different customers for the same product based on segment, time, or quantity (e.g., student discounts, peak/off-peak electricity, airline ticket pricing).

📄 Practice Model Paper

BASED ON OFFICIAL PATTERN

⚠️ **Disclaimer:** This is a practice model paper for study purposes only. It is **not** an official SITTTR model question paper. The pattern follows the official ESE structure (2.5 hours, 60 marks, Parts A, B, C) as described in the syllabus.

Course: Managerial Economics (1143) · Semester I · Practicum
Duration: 2.5 hours · Total Marks: 60

Part A (8 × 1 = 8 marks)

Answer all questions. Each carries 1 mark.

1. Define Managerial Economics.
2. State the Law of Diminishing Marginal Utility.
3. What is the Law of Demand?
4. Define price elasticity of demand.
5. What is a production function?
6. What are economies of scale?
7. Name any two forms of market.
8. What is Cost Plus Pricing?

Part B (4 × 3 = 12 marks)

Answer any 2 from each module. Each carries 3 marks.

Module I

1. Explain the concept of utility and its types.
2. What are the factors that affect demand?
3. Distinguish between value and price.

Module II

1. Explain the exceptions to the Law of Demand.
2. What are the types of elasticity of demand?
3. Distinguish between the Law of Demand and the Law of Supply.

Module III

1. Explain the Law of Variable Proportion.
2. What are the factors of production?
3. Distinguish between fixed cost and variable cost.

Module IV

1. Describe the features of perfect competition.
2. What are the types of monopoly?
3. Explain differential pricing with an example.

Part C (4 × 7 = 28 marks)

Answer one set with choice from each module. Each carries 7 marks.

Module I

Set 1: Define Managerial Economics. Explain its scope and importance in business decision-making. *OR* **Set 2:** Explain the Law of Diminishing Marginal Utility. Illustrate with a table and diagram.

Module II

Set 1: Explain the concept of elasticity of demand. Discuss its types and factors affecting it. *OR* **Set 2:** Explain the Law of Supply and its exceptions. Discuss the concept of elasticity of supply.

Module III

Set 1: Explain the production function. Discuss the Law of Variable Proportion with its three stages. *OR* **Set 2:** Explain the different cost curves (ATC, AVC, MC) with a diagram. Discuss economies and diseconomies of scale.

Module IV

Set 1: Compare and contrast perfect competition, monopoly, and monopolistic competition. *OR* **Set 2:** Explain the various methods of pricing used in different market structures.

⚡ Quick Revision EXAM PREP CARDS

📘 Module I — Introduction

- **Managerial Economics:** Economic theory applied to business decisions.
- **Utility:** Satisfaction from consumption. TU = Σ MU.
- **Law of DMU:** MU declines with each additional unit.
- **Value vs. Price:** Value = worth; Price = monetary expression.

📗 Module II — Demand & Supply

- **Law of Demand:** P ↑ → Qd ↓ (inverse).
- **Law of Supply:** P ↑ → Qs ↑ (direct).
- **Elasticity:** Ed = %ΔQd / %ΔP. Types: elastic, inelastic, unit.
- **Exceptions:** Giffen goods, Veblen goods, speculation.

📙 Module III — Production

- **Production function:** Q = f(L, K, Land, E).
- **LVP:** 3 stages — Increasing, Diminishing, Negative returns.
- **Cost curves:** FC (horizontal), VC (upward), TC (parallel), ATC/AVC/MC (U-shaped).
- **Economies of scale:** ↓ AC as output ↑; Diseconomies: ↑ AC as output ↑.

📕 Module IV — Market Forms

- **Perfect competition:** Many firms, homogeneous, price-takers.
- **Monopoly:** One firm, unique product, price-maker.
- **Monopolistic competition:** Many firms, differentiated products.
- **Oligopoly/Duopoly:** Few firms, interdependent decisions.
- **Pricing methods:** Cost-plus, target, marginal cost, going rate, customary, differential.

⚠️ Common Mistakes to Avoid

- **Confusing demand with quantity demanded:** Demand is the entire curve; quantity demanded is a point on it.
- **Misapplying elasticity formulas:** Always use the percentage change formula correctly. Watch the sign — price elasticity is usually negative.

- **Mixing up fixed and variable costs:** Fixed costs do not change with output; variable costs do.
- **Confusing market structures:** Remember the number of firms, product type, and barriers to entry for each.
- **Forgetting ceteris paribus:** Economic laws assume "other things being equal" — don't ignore this condition.

✅ Exam Checklist

- 🔲 Definitions of Economics and Managerial Economics
- 🔲 Law of DMU — table and diagram
- 🔲 Law of Demand — curve, exceptions, reasons for downward slope
- 🔲 Elasticity of demand — types, calculation, factors
- 🔲 Law of Supply — curve, exceptions
- 🔲 Production function and Law of Variable Proportion
- 🔲 Cost curves — draw and label all
- 🔲 Economies and diseconomies of scale
- 🔲 Market forms — features of each
- 🔲 Pricing methods — examples of each

📖 Glossary KEY TERMS

Term	Meaning
Managerial Economics	Application of economic theory to business decision-making
Utility	Satisfaction from consuming a good
Marginal Utility	Additional utility from one more unit
Law of Diminishing Marginal Utility	MU declines as consumption increases
Demand	Willingness and ability to buy at a given price
Supply	Willingness and ability to sell at a given price
Elasticity of Demand	Responsiveness of Qd to price/income/other price changes
Elasticity of Supply	Responsiveness of Qs to price changes
Production Function	Relationship between inputs and output
Law of Variable Proportion	MP of a variable input eventually declines
Returns to Scale	Output response when all inputs change proportionally
Fixed Cost	Cost that does not change with output
Variable Cost	Cost that changes with output
Marginal Cost	Additional cost of one more unit
Economies of Scale	Cost advantages from larger production
Diseconomies of Scale	Cost disadvantages from excessive production

Term	Meaning
Perfect Competition	Market with many firms, homogeneous goods, price-takers
Monopoly	Market with a single seller, unique product, price-maker
Monopolistic Competition	Many firms, differentiated products, some price control
Oligopoly	Few firms, interdependent decisions, high barriers
Duopoly	Two firms dominate the market
Cost Plus Pricing	Price = cost + mark-up
Differential Pricing	Charging different prices to different customers

📚 References

TEXTBOOKS & RESOURCES

📖 Textbooks

- **Managerial Economics** — Dr. S. Sankaran, Margham Publications, Chennai.
- **Managerial Economics** — Dr. P.C. Thomas, Kalyani Publications.

🌐 Web Resources

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- (Additional NPTEL / online resources as recommended by faculty.)

Major Equipment (for batch of 30 students)

Particulars	Specifications	Quantity
Computers with internet facility	64 Mbps (Minimum)	30
MS Office packages	Latest	30